

Employee

1. Feature Overview

ADMIN management of staff users (employees): list, create, update, including roles used by Spring Security.

Status: Implemented (backend package name `user` ; UI label Employees)

2. Business Purpose

Provision bank staff who can log in and operate customer/account functions according to role.

3. User Workflow

1. Sidebar → Employees (`/app/employees`) — ADMIN only
2. List / create / edit via dialogs
3. Assigned roles control API and route access

4. Execution Flow

```
UserController.*
      ↓
UserService.*
      ↓
UserRepository / UserRoleRepository / RoleRepository
```

5. Database Tables

`users` , `roles` , `user_roles`

6. REST APIs

Method	Path	Roles
GET	<code>/users</code>	ADMIN
POST	<code>/users</code>	ADMIN
PUT	<code>/users/{id}</code>	ADMIN

7. Controllers

`com.bankone.user.controller.UserController`

8. Services

`UserService` (+ `AdminRoleInitializer` for bootstrap admin)

9. Repositories

`UserRepository` , `UserRoleRepository` , `RoleRepository`

10. DTOs

Request/response DTOs under `com.bankone.user.dto` (as present in codebase)

11. Entities

`User` , `Role` , `UserRole`

12. Utility Classes

`EmployeeSpecification` for search

13. Configuration Classes

`SecurityConfig` `/users/**` → `hasRole("ADMIN")`

`RoleInitializer` seeds role names including unused TELLER, AUDITOR, CUSTOMER

14. Validation Rules

Password encoding BCrypt on create/update; field validation per DTOs

15. Security Rules

ADMIN only for all `/users` APIs and Angular route

16. Exception Handling

Duplicate username / missing role → business exceptions

17. Logging

Security TRACE may log auth for employees in dev

18. Audit Events

`AuditableEntity` fields on user/role join entities

19. Testing Strategy

- Non-ADMIN gets 403 on `/users`
- Create employee → login works
- Disable user → login blocked

20. Future Extension Guide

- Self-service profile already under auth; keep employee admin separate
- Activate TELLER/AUDITOR in `SecurityConfig` when features land

- Branch assignment when Branch module exists

Future Modification Guide

Requirement: Allow MANAGER to list employees (read-only)

Item	Detail
Files	SecurityConfig.java , UserController (optional @PreAuthorize), app.routes.ts , sidebar
Methods	securityFilterChain matchers for GET /users/**
Impact	Data exposure review

Requirement: Add mandatory employee code format

Item	Detail
Files	User entity/DTO, UserService create/update, Angular employee dialog
Impact	Login still uses username; display uses employeeCode

Call hierarchy (create)

```
UserController.create...
    ↓
UserService.create...
    ↓
PasswordEncoder.encode()
    ↓
UserRepository.save()
    ↓
UserRoleRepository.save()
```